Polynomials — Solutions

Problem 1. Let a, b, c, d be real numbers which are not all zero. Prove that the roots of the polynomial $x^6 + ax^3 + bx^2 + cx + d$ cannot all be real numbers.

Solution. Let $\alpha_1, \alpha_2, \dots, \alpha_6$ be the roots of the polynomial. By Vieta's theorem, we have

$$\sum_{i=1}^{6} \alpha_i = 0 \quad \text{and} \quad \sum_{1 \leqslant i < j \leqslant 6} \alpha_i \alpha_j = 0.$$

It follows that

$$\sum_{i=1}^{6} \alpha_i^2 = \left(\sum_{i=1}^{6} \alpha_i \right)^2 - 2 \sum_{1 \leqslant i < j \leqslant 6} \alpha_i \alpha_j = 0.$$

If all roots are real, then we must have $\alpha_1 = \alpha_2 = \cdots = \alpha_6 = 0$. This implies $a = b = c = d = 0$, contradiction. $\square$

Problem 2. Find all polynomials $P(x)$, $Q(x)$ and $R(x)$ such that

$$P(x) - Q(x) = R(x) \left(\sqrt{P(x)} + \sqrt{Q(x)} \right).$$

Solution. If P and Q are the zero polynomial, then R can be any polynomial. If $P = Q \neq 0$, then R must be the zero polynomial.

Suppose not both of P and Q are the zero polynomial and $P \neq Q$. We divide both sides of the given equation by $\sqrt{P(x)} + \sqrt{Q(x)}$ to obtain

$$\sqrt{P(x)} - \sqrt{Q(x)} = R(x). \tag{1}$$

Squaring both sides, we see that $P(x) + Q(x) - 2\sqrt{P(x)Q(x)} = R(x)$. This shows $\sqrt{P(x)Q(x)}$ is a polynomial. Thus, we can write $P(x) = f(x)^2 h(x)$ and $Q(x) = g(x)^2 h(x)$ for some real polynomials f, g, h such that h has no nontrivial square factors. Equation (1) becomes

$$(|f(x)| - |g(x)|) \sqrt{h(x)} = R(x).$$

As $P \neq Q$, this shows $\sqrt{h(x)}$ is a polynomial, and hence h is a constant polynomial. Since $h(x) \geqslant 0$, we may assume $h(x) = 1$. Then $|f(x)| - |g(x)| = R(x)$, which means

$$\pm f(x) \pm g(x) = R(x).$$

There are infinitely many x that correspond to the same choices of the signs. This gives a polynomial equation satisfied by infinitely many x, and hence it is an identity. In other words, both f and g have constant signs. WLOG we may assume both of them are nonnegative. Therefore, in this case, $P(x) = f(x)^2$, $Q(x) = g(x)^2$ and $R(x) = f(x) - g(x)$ where f and g are nonnegative polynomials. $\square$

Problem 3. Find all polynomials $P(x)$ with integer coefficients such that $P(n) \mid 2^n + 3^n$ for all positive integers n.

Solution. If P is a constant polynomial, we must have $P \mid 2^1 + 3^1 = 5$ and $P \mid 2^2 + 3^2 = 13$. This shows $P(x) = \pm 1$, both are obviously solutions.

If P is not a constant polynomial, there must be $m \in \mathbb{N}$ such that $P(m) \neq \pm 1$. Let p be a prime divisor of $P(m)$. Note that $p > 3$ since $p \mid P(m) \mid 2^m + 3^m$. By the Chinese remainder theorem, there exists a positive integer n such that

$$\begin{aligned} n &\equiv m \pmod{p}, \\ n &\equiv 1 \pmod{p-1}. \end{aligned}$$

As $n \equiv m \pmod{p}$, we have $P(n) \equiv P(m) \equiv 0 \pmod{p}$. Thus, $p \mid 2^n + 3^n$. By the Fermat little theorem, we have

$$0 \equiv 2^n + 3^n \equiv 2^1 + 3^1 = 5 \pmod{p}.$$

This implies $p = 5$. Similarly, there exists a positive integer k such that

$$\begin{aligned} k &\equiv m \pmod{p}, \\ k &\equiv 2 \pmod{p-1}. \end{aligned}$$

Using the same argument, we find that $0 \equiv 2^2 + 3^2 = 13 \pmod{p}$, contradicting $p = 5$. Therefore, the only polynomials are $P(x) = \pm 1$. $\square$

Problem 4. Find all polynomials $P(x)$ with real coefficients satisfying the following condition: For any real number x, $P(x)$ is an integer if and only if x is an integer.

Solution. Note that $P(x)$ cannot be a constant polynomial. WLOG assume the leading coefficient of $P(x)$ is positive. Thus, there exists N such that $P(x)$ is strictly increasing for all $x > N$. For any integer $m > N$, as $P(m+1)$ and $P(m)$ are integers, we must have $P(m+1) \geqslant P(m) + 1$. If $P(m+1) \geqslant P(m) + 2$, then there must be some $c \in (m, m+1)$ such that $P(c) = P(m) + 1$ by the intermediate value theorem. This is a contradiction. Therefore, we must have

$$P(m+1) = P(m) + 1$$

for all integers $m > N$. But since this holds for infinitely many m, we must have $P(x+1) = P(x) + 1$ for all $x \in \mathbb{R}$. This easily implies $P(x) = x + a$. Clearly, the only solutions are those polynomials $x + a$ with $a \in \mathbb{Z}$. $\square$

Problem 5. Find all polynomials $P(x)$ with integer coefficients satisfying the following condition: There exists an arithmetic progression $\{a + nd\}_{n \in \mathbb{Z}}$ such that $P(x) \neq a + nd$ for any integers x and n.

Solution. We consider those $P(x)$ which do not satisfy the given condition. For every $d \in \mathbb{N}$, by considering all arithmetic progressions with common difference d, we know that $P(x) \equiv a \pmod d$ is solvable for every $a \in \mathbb{Z}$. If we regard P as a function from $\{0, 1, \ldots, d-1\}$ to itself modulo d, then P is surjective. Therefore, P is injective. In other words, whenever $P(x) \equiv P(y) \pmod d$, we have $x \equiv y \pmod d$.

Now, consider any x and y such that $P(x) \neq P(y)$, we can take $d = |P(x) - P(y)| > 0$. Then $P(x) \equiv P(y) \pmod d$, and hence $x \equiv y \pmod d$ from above. This implies $P(x) - P(y) \mid x - y$. In particular, $P(y+1) - P(y) \mid 1$, which implies $P(y+1) - P(y) = \pm 1$ for all y. As P is a polynomial, either $P(y+1) - P(y) = 1$ for all y or $P(y+1) - P(y) = -1$ for all y. It follows that $P(x) = \pm x + c$. In either case, the range of P is $\mathbb{R}$. It is clear that P does not satisfy the condition.

Therefore, P can be any polynomial different from $\pm x + c$. □

Problem 6. (Turkey TST 1995 Problem 1) Let a and b be real numbers satisfying $b \geqslant a > 0$. Find all real numbers $x_1, x_2, \ldots, x_n$ (in terms of a and b) such that $x_j^2 + 2ax_j + b^2 = x_{j+1}$ for $j = 1, 2, \ldots, n$, where $x_{n+1} = x_1$.

Solution. Firstly, we have

$$x_{j+1} = x_j^2 + 2ax_j + b^2 = (x_j + a)^2 + (b^2 - a^2) \geqslant 0.$$

As $a, b > 0$, $f(x) = x^2 + 2ax + b^2$ is increasing. WLOG assume $x_1 = \max\{x_1, x_2, \ldots, x_n\}$. As $x_1 \geqslant x_2$, we have $x_2 = f(x_1) \geqslant f(x_2) = x_3$. Inductively, we obtain

$$x_1 \geqslant x_2 \geqslant x_3 \geqslant \cdots \geqslant x_n \geqslant x_1.$$

Thus, all x_j's are equal. Then all equations become $x_1^2 + 2ax_1 + b^2 = x_1$. The only solution is

$$x_1 = x_2 = \cdots = x_n = \frac{-(2a-1) \pm \sqrt{(2a-1)^2 - 4b^2}}{2}. \tag{2}$$

Note that we need $(2a-1)^2 - 4b^2 \geqslant 0$, which means $1 - 2a \geqslant 2b$ (it is impossible that $2a - 1 \geqslant 2b$ as $b \geqslant a$). In other words, if $a + b \leqslant \frac{1}{2}$, then (2) gives the only solutions. Otherwise, there is no real solution. □

Problem 7. Find all polynomials $P(x)$ with integer coefficients such that $P(n)$ is a palindrome number for every positive integer n. (A palindrome number is a positive integer that reads the same from left to right and from right to left.)

Solution. Clearly, $P(x)$ can be any constant polynomial such that the constant is a palindrome number. Assume $\deg P \geqslant 1$. Since $P(n)$ is a positive integer for all $n \in \mathbb{N}$, the leading coefficient of P must be positive. We claim that there are no such polynomials.

Suppose the unit digit of $P(1)$ is a. As $P(10m+1) \equiv P(1) \pmod{10}$ for any integer m, the unit digit of $P(10m+1)$ is a. Thus, the first digit of $P(10m+1)$ is a.

Since P is not bounded above, there must be infinitely many m such that $P(10(m+1)+1)$ has more digits than $P(10m+1)$. From above, we must have

$$\frac{P(10(m+1)+1)}{P(10m+1)} > \frac{a \cdot 10^{k+1}}{(a+1) \cdot 10^k} = \frac{10a}{a+1} > 2. \tag{3}$$

However, if the leading term of P is $a_d x^d$, then we have

$$\lim_{x\to\infty} \frac{P(10(x+1)+1)}{P(10x+1)} = \lim_{x\to\infty} \frac{a_d(10x)^d}{a_d(10x)^d} = 1.$$

Therefore, (3) fails to hold for sufficiently large m, contradiction. $\square$

Problem 8. (China TST 2017 Test 1 Problem 4) Find all pairs (m, n) of positive integers such that there exist two monic polynomials $P(x)$ and $Q(x)$ with $\deg P = m$ and $\deg Q = n$ satisfying $P(Q(t)) \neq Q(P(t))$ for all real numbers t.

Solution. (m, n) can be any pair of positive integers except $(1, 1), (1, 2k), (2k, 1)$ for some $k \in \mathbb{N}$.

If $m = n = 1$, let $P(x) = x + a$ and $Q(x) = x + b$. Then

$$P(Q(x)) - Q(P(x)) = P(x+b) - Q(x+a) = (x+b+a) - (x+a+b) = 0.$$

If $(m, n) = (1, 2k)$ for some $k \in \mathbb{N}$ (or similarly $(m, n) = (2k, 1)$), we let $P(x) = x + a$ and $Q(x) = x^{2k} + b_{2k-1}x^{2k-1} + \cdots + b_1 x + b_0$. Then

$$P(Q(x)) - Q(P(x)) = (x^{2k} + b_{2k-1}x^{2k-1} + \cdots) - \big((x+a)^{2k} + b_{2k-1}(x+a)^{2k-1} + \cdots\big),$$

where $\cdots$ contains terms of smaller degrees. This shows the leading term of $P(Q(x)) - Q(P(x))$ is $2kax^{2k-1}$. This means it is a polynomial with odd degree unless $a = 0$, which means it must have a root. But even if $a = 0$, we still have $P(Q(x)) - Q(P(x)) = 0$.

If $m > 1$ and n is even (or similarly m is even and $n > 1$), we choose $P(x) = x^m$ and $Q(x) = x^n + 2$. Then

$$P(Q(x)) - Q(P(x)) = (x^n+2)^m - (x^{mn}+2) = \left(\sum_{j=1}^{m-1} \binom{m}{j} 2^{m-j} x^{jn}\right) + (2^m - 2).$$

As n is even, all monomials have even degrees. Also, the coefficients $\binom{m}{j}2^{m-j}$ and $2^m - 2$ are positive. This shows $P(Q(x)) - Q(P(x)) > 0$ for all x.

If both m, n are odd and $m \geqslant 3$ (or similarly $n \geqslant 3$), we choose $P(x) = x^m$ and $Q(x) = x^n + 4$. Then

$$P(Q(x)) - Q(P(x)) = (x^n+4)^m - (x^{mn}+4).$$

If we can prove that the polynomial

$$R(y) = (y+4)^m - (y^m + 4)$$

has no real root, then it is clear that $P(Q(x)) - Q(P(x))$ has no real root (otherwise we take $y = x^n$).

Note that the leading term of $R(y)$ is $4my^{m-1}$, which has an even degree and a positive coefficient. Thus, R has a minimum value. Now,

$$R'(y) = m(y+4)^{m-1} - my^{m-1}.$$

As $m \geqslant 3$ is odd, we have $R'(y) = 0$ if and only if $y + 4 = \pm y$. Clearly, the only possibility is $y + 4 = -y$, which means $y = -2$. Since this is the only critical point, R must attain its minimum value at $y = -2$. It is easy to check that

$$R(-2) = 2^m - (-2)^m - 4 = 2^{m+1} - 4 > 0.$$

Therefore, R is always positive and it has no real root. $\square$